

Loop-factor library $\mathcal{L}$

Each row is one parameter-free class of the form $1 \pm c_\sigma$, derived from a Lattice-Defect-FT spinor-trace lemma.

Lemma	Class name	Loop class L_σ	Physical origin
1	Yukawa-Damping	$1 \pm \gamma/4$	$d = 4$ spinor-trace normalization
2	PMNS-Self-Energy	$1 \pm \gamma^2/4$	double-Wick contraction
3	Pure-Self-Energy	$1 \pm \gamma^2$	spinor-trace-less vertex
4	Resummed-Propagator	$1/(1 \pm 2\gamma^2)$	resummation across two chiralities
5	Generation	$1 \pm \gamma/N_{\text{gen}}$	generation-summed self-energy
6	Sub-Generation	$1 \pm \gamma/(2N_{\text{gen}})$	$SU(2)_L$ doublet/singlet splitting
7	EW-Mixed	$1 \pm \gamma \epsilon_{\text{sync}}^2$	bosonic Lemma-1 with Goldstone vertex
8	Cosmological-Density	$1 \pm \gamma/2$	chirality restriction (matter-only)
--	Pure-Sync	$1 \pm \epsilon_{\text{sync}}^2$	pure Goldstone-vertex bosonic class